Which Isolates Should the Laboratory Test? Active Learning, Drift and Calibration for MALDI-TOF Antimicrobial Resistance Prediction

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Abstract

Models that read antibiotic resistance directly from MALDI-TOF mass spectra are attractive because the spectrum is already acquired during routine identification, but their accuracy falls away as bacterial populations and laboratory conditions change, and keeping them useful means retraining them on isolates that have been freshly phenotyped. Phenotyping is the expensive step. A laboratory that can pay for only a limited number of susceptibility tests in a year therefore has to decide which of the arriving isolates to send, and as far as we are aware this question has not been examined for this modality. We report 2 859 retraining simulations on DRIAMS covering three organism and antibiotic pairs, four hospitals, label budgets between 40 and 800, and ten random seeds. Random selection is compared against uncertainty, diversity and hybrid acquisition, against conformal safety routing, against pseudo labelling, and against recalibration of the deployed decision rule. None of the acquisition strategies produced a better model than random sampling in any of the 45 preplanned comparisons, and the primary endpoint that we had specified in advance, the number of labels needed to recover 95 % of the gain from full retraining, was not reached within 800 labels by any strategy. The choice of rule still matters, however, since an isolate that is tested receives a laboratory answer rather than a model answer. Acquiring the isolates the model was least sure about reduced the resistant isolates that were wrongly reported susceptible during the year by 33 to 73 % ($p < 0.01$ for all three pairs), although the effect on the opposite error turned out to depend on the organism. A third result concerns the safety machinery itself. A decision threshold calibrated for 90 % sensitivity was delivering 73 to 80 % two years later, and a conformal rule that promised at most 10 % miscoverage on resistant isolates delivered 14 to 19 %, in neither case with any warning sign in the discrimination metric. Recalibrating on freshly acquired labels brought the sensitivity target back, but it roughly doubled the opposite error, which is what 90 % sensitivity actually costs at an AUROC between 0.75 and 0.85. A preregistered search over 16 method combinations, run on a development split that never touches the test year, gave the same picture and found no combination that improved both error types beyond uncertainty acquisition on its own. We argue from this that label efficiency is the wrong objective for the problem, and that acquisition policies should be reported with counts of both error types at deployment, with calibration treated as a component that has to be maintained rather than as a property of the training run.

Keywords: Antimicrobial resistance, MALDI-TOF mass spectrometry, Active learning, Dataset shift, Conformal prediction, Model calibration, Clinical decision support

1 Introduction

In routine practice, antimicrobial resistance is established by phenotypic susceptibility testing, and the result takes a day or more to come back once the organism has been identified. Identification itself is almost immediate, because clinical laboratories already acquire a MALDI-TOF mass spectrum of every isolate as part of that step. Weis et al. [1] showed that those same spectra carry a usable resistance signal, and released DRIAMS, a collection of spectra from four hospitals paired with laboratory susceptibility labels. Areas under the receiver operating characteristic curve of approximately 0.80, 0.74 and 0.74 were reported for *Escherichia coli*, *Klebsiella pneumoniae* and *Staphylococcus aureus* on clinically important antibiotics. A model that reads the spectrum can therefore offer a provisional resistance call several hours before the phenotype is available.

Such a model is not a static artefact. Bacterial populations change, and so do laboratory workflows and instruments, and the same study documented that predictive accuracy decays over time. Keeping the model useful therefore means retraining it on recently phenotyped isolates, and phenotyping is precisely the cost that the model was meant to avoid in the first place. A laboratory that can afford, say, four hundred extra susceptibility tests in a year is then left with a concrete operational question: *which* four hundred of the several thousand arriving isolates should be tested?

This is the classical pool based active learning problem [2], and the natural expectation is that a well chosen subset buys more accuracy per test than an arbitrary one. Our aim was to measure how much more. Over 2 859 simulated retraining runs we found no measurable advantage at all. The negative result seems to us worth reporting in its own right, since the opposite is widely assumed in applied work. It is not the whole story, though, and much of what follows is concerned with what that result hides.

The acquisition rule decides more than what the model learns next. It also decides *who receives a laboratory answer instead of a model answer.* An isolate that is sent for testing is, from the patient's point of view, protected, because its true phenotype becomes known before treatment decisions are finalised. An isolate that is not tested is reported with the model's call, errors and all. Acquisition is in that sense a patient safety decision taken while the year is running, and it is largely separate from what the same decision does for next year's model. On that axis the strategies do differ, and they differ sharply.

A third thread runs through the results. The decision threshold and the conformal prediction sets used in this setting are both calibrated on historical data, and both make explicit numerical promises: 90 % sensitivity to resistance in the first case, and at most 10 % miscoverage per class in the second. Under drift the two promises failed without any obvious symptom, while the headline discrimination metric remained healthy. We measure how large the failure becomes, repair it by recalibrating on newly acquired labels, and show that the repair uncovers an operating cost which the broken calibration had been hiding.

The contributions of this work can be summarised as follows.

- To our knowledge this is the first study of active isolate acquisition for the maintenance of MALDI-TOF resistance models. It comprises six experiments and 2 859 simulated retraining runs over three organism and antibiotic pairs and four hospitals, with full seed replication and paired statistical testing.
- We report a clearly stated negative result on label efficiency, together with the statistical power that supports it, and we set it beside a positive result at deployment concerning the errors that actually reach patients.
- We quantify how far threshold calibration and conformal calibration decay under temporal drift, and measure what restoring the sensitivity promise costs in terms of the opposite error.
- We carry out a preregistered search over sixteen method variants on a development split that never touches the test year, which reproduces our conclusions and eliminates the obvious remedies.
- We give an explicit account of a design flaw in our own semisupervised experiment, which is reported rather than quietly discarded, so that the corresponding question is left open instead of answered incorrectly.

The rest of the paper is organised as follows. Section 2 reviews related work on resistance prediction from mass spectra, on active learning and on calibration under shift. Section 3 formalises the maintenance loop, the error taxonomy and the acquisition strategies. Section 4 describes the data, the experimental matrix and the statistical protocol. Section 5 reports the results. Section 6 discusses what they imply, Section 7 states the limitations and Section 8 concludes.

2 Related Work

In this section we summarise the relevant progress in resistance prediction from mass spectra, in uncertainty quantification for that modality, in active learning and its behaviour under distribution shift, and in the maintenance of calibration, and we identify the gap that motivates the present study.

2.1 Resistance Prediction from MALDI-TOF Spectra

DRIAMS established the task, the standard preprocessing and the reference baselines. It assembles more than 300 000 spectra with laboratory confirmed resistance profiles collected at four Swiss diagnostic laboratories [1]. Later work has extended the approach to further organisms and to improved spectral representations. The temporal decay of these models is documented within the same line of work, so

demonstrating decay is not a contribution of the present paper. It is the premise on which our study rests, and we verify it once more under our own protocol in Section 5.1.

2.2 Uncertainty Quantification for This Modality

A classifier intended for clinical microbiology has to be able to decline a prediction when it cannot make one reliably. Corvelo Benz et al. [3] applied conformal prediction to MALDI-TOF resistance classification and obtained prediction sets carrying distribution free statistical guarantees, while earlier work had introduced kernels tailored to mass spectra together with well calibrated uncertainty estimates [4]. Our use of conformal sets as an *acquisition* rule, that is, as a way of testing exactly those isolates that the model cannot resolve, builds directly on that work. What we add to it is the observation that the guarantee does not survive a year of drift unless it is recalibrated, along with a measurement of how large the violation becomes.

2.3 Active Learning and Query Strategies

Pool based active learning requests expert annotation for the most informative unlabelled samples in an iterative fashion [2]. Uncertainty sampling [5], core set and k-center diversity [6], and hybrids of the two make up the standard families. Importance weighted [7], bandit selected [8] and cost sensitive [9] variants have been proposed to deal with sampling bias, strategy selection and asymmetric misclassification cost respectively. It has also been observed repeatedly in benchmark studies that random acquisition is a stubborn baseline, and that deliberately injecting randomness into uncertainty sampling works as a stabiliser [10]. In medical imaging the annotation burden has been reduced substantially by active learning, even though the evaluation in those studies is almost always label efficiency on a held out set rather than the errors that are incurred while the loop is running.

2.4 Active Learning under Distribution Shift

Malware detection has the same structure as our problem, namely a deployed classifier, a continuous arrival of new samples, a bounded analyst budget and continuous drift. Label efficient update policies [11] and the decay of conformal evaluation under drift [12] are both documented in that setting. The active selective prediction setting combines acquisition with the option to abstain and hand the case to a human [13]. None of this literature deals with the question of which bacterial isolates a clinical laboratory should phenotype, and as far as we are aware none of it uses the random share of an active learning budget as the calibration set for the deployed decision rule.

2.5 Calibration under Shift

Weighted conformal prediction restores coverage under covariate shift provided that a density ratio can be estimated [14], and adaptive conformal inference tracks shift online by adjusting the target level in response to observed feedback [15]. Both are implemented as candidate remedies in the combination search of Section 5.7. The underlying split conformal framework is standard [16].

3 Problem Setting and Method

3.1 The Maintenance Loop

We simulate the loop that a laboratory would actually run, which is shown in Fig. 1. A model trained on historical isolates is in service. New isolates arrive through the year in $R = 4$ rounds. In round r the model currently in service scores every arriving isolate, an acquisition rule spends b/R of a fixed annual budget b on susceptibility tests, the tested isolates come back with their true labels, and the model together with its decision threshold and its conformal bounds is refitted on everything that has been acquired so far. Isolates that were not tested are reported with the model's call, and an error there is an error a clinician may act on. At the end of the year the model is evaluated on the following year's isolates, which are never visible while the loop is running.

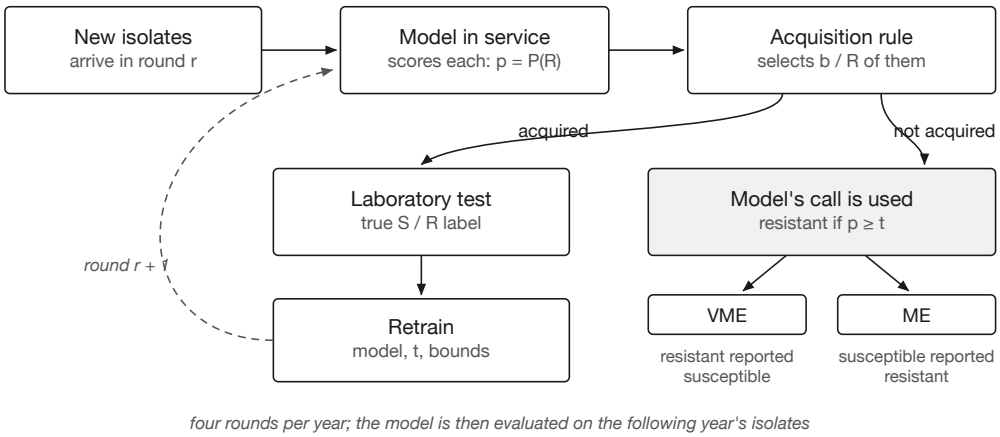


Fig. 1: The loop that every run simulates. The acquisition rule determines both what the model learns next (upper path) and which patients receive a laboratory answer rather than a model answer (lower path). Errors on the lower path are counted as they occur, not only at the end of the year. VME and ME are defined in Section 3.2.

3.2 Error Taxonomy and Reported Quantities

Two error types are distinguished throughout the paper, following the terminology that is used to evaluate susceptibility testing devices [17]. A **very major error** (VME) is a resistant isolate that is reported as susceptible, so the patient may be given an antibiotic that does not work. A **major error** (ME) is a susceptible isolate reported as resistant, so the patient may be given an unnecessarily broad antibiotic, which carries its own harm and its own cost. The two are not interchangeable. A method that lowers one of them while raising the other has not made anything safer; it has only moved along a tradeoff.

We report both error types in two different views, and this distinction is central to the paper. The *predictive* view gives rates on the following year's isolates and measures how good the retrained model is. The *operational* view gives counts among the untested isolates of the current year and measures how many patients were actually affected while the loop was running. Writing U_r for the isolates of round r that the rule did not acquire, the operational very major error count is

$$\mathrm{VME}_{\mathrm{op}} = \Sigma_{\mathrm{r}=1..\mathrm{R}} \, | \, \{ \, \mathrm{i} \in \mathrm{U}_{\mathrm{r}} : \mathrm{y}_{\mathrm{i}} = \mathrm{R} \text{ and } \mathrm{p}_{\mathrm{i}} < \mathrm{t}_{\mathrm{r}} \, \} \, |,$$

where y_i is the true phenotype, p_i the probability that was assigned by the model in service when isolate i arrived, and t_r the threshold in force at that moment; $\mathrm{ME}_{\mathrm{op}}$ is defined symmetrically. Since the rule and the model both change between rounds, these counts cannot be recovered from metrics that are computed at the end of the year, and it is on these counts that the acquisition strategies separate.

3.3 Acquisition Strategies

Let p be the model's probability of resistance for an arriving isolate and t the current decision threshold. In each round the rule selects b/R isolates out of that round's arrivals.

- **Random.** Uniform selection without replacement; this is the control condition.
- **Uncertainty.** Smallest $|p - t|$, that is, the isolates whose call is closest to flipping [5].
- **Diversity.** Greedy k-center selection in a 50-dimensional principal component space, which maximises coverage of spectrum space relative to what has already been labelled [6].
- **Hybrid.** The $4b$ most uncertain isolates are clustered by k-means into b groups, and the isolate nearest each centroid is acquired.
- **Conformal routing.** Only isolates whose per class prediction set is ambiguous, that is $\{\mathrm{S}, \mathrm{R}\}$ or empty, are acquired [3]. We report this both capped at the budget and uncapped.
- **Bounds.** *No retraining* (budget zero) and *full retraining* (every arriving isolate labelled) bracket what any rule can achieve.

3.4 Decision Threshold and Conformal Bounds

The operating threshold is taken to be the largest t that achieves a target sensitivity $\tau = 0.90$ to resistance on held out data, since that is the quantity a laboratory would specify. Mondrian (per class) split conformal prediction supplies prediction sets at $\alpha = 0.10$, calibrated separately on the susceptible and the resistant validation isolates so that the minority class receives a guarantee of its own [16]. In the baseline configuration both are fitted once on the 2015 and 2016 validation split and then left untouched, which is what a deployed system does by default; Section 5.5 reports what happens when they are instead refitted each round on freshly acquired labels.

4 Experimental Setup

4.1 Dataset and Splits

All of the experiments use DRIAMS [1]. Each isolate is represented by one MALDI-TOF spectrum binned into 6 000 intensity features and paired with a laboratory susceptibility label. Intermediate and resistant are both treated as resistant, following the reference implementation that was released with the dataset. We study the three organism and antibiotic pairs that have enough labelled isolates for repeated simulation.

The temporal protocol is shown in Fig. 2. It trains on the 2015 and 2016 isolates of hospital DRIAMS-A, holding out 20 % as the validation split that is used for the threshold and the conformal bounds, spends the label budget on 2017 isolates arriving in four rounds, and evaluates on 2018. DRIAMS does not record acquisition dates, so the four arrival rounds are a fixed pseudo random partition of the 2017 isolates derived from a hash of the isolate identifier. This reproduces the structure of a budget that is spent over time, but it *cannot* reproduce drift within the year, a limitation we return to in Section 7. The transfer protocol moves the model trained on 2015 and 2016 to hospitals B, C and D, and allows each of them to acquire local labels from its own 2018 isolates, holding out a stratified test split at each site.

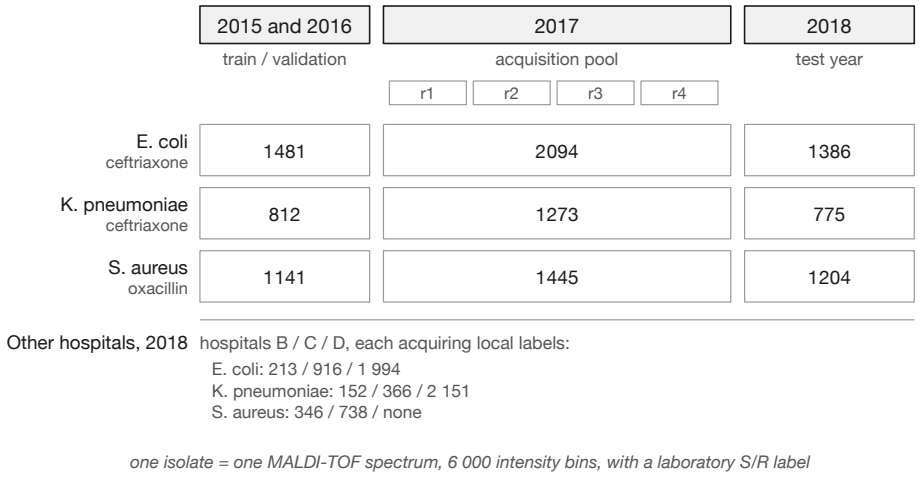


Fig. 2: Data and splits. Counts are isolates per organism and antibiotic pair. The 2018 isolates of hospital A are reserved for final evaluation, and the development split used for the combination search of Section 5.7 never touches them. *S. aureus* has no labelled oxacillin isolates at hospital D, so that combination is not run.

4.2 Experimental Matrix

Six experiments were executed, and they are summarised in Table 1. Every configuration is replicated across seeds, and the data split, the hyperparameters and the random seed are held fixed within a comparison, so that any difference can be attributed to the acquisition rule and the calibration policy alone.

Table 1: Experimental matrix. Budgets are total labels acquired over the year, divided evenly across the four arrival rounds. Runs include the no retraining and full retraining bounds where applicable. One E5 run was lost in transfer, giving 239 of 240.

Id	Question	Acquisition rules	Budgets	Seeds	Runs
E1	Label efficiency and acquisition family	random, uncertainty, diversity, hybrid	40 to 800	10	660
E2	Value of unlabelled spectra	the above + pseudo labelling	100 to 800	10	390
E3	Conformal routing as an acquisition rule	routing, routing (uncapped)	40 to 800	10	180
E4	Adoption at a different hospital	random, uncertainty, diversity, hybrid	25 to 400	5	880
E5	Recalibration on acquired labels	random, uncertainty	100 to 800	10	239
D1	Preregistered combination search	16 combinations (Section 5.7)	100, 300	5	510
Total					**2 859**

4.3 Protocol Constants

Table 2 lists the quantities that are held constant across all runs. Hyperparameters were tuned once per organism and antibiotic pair on the training years by cross validation and then frozen, so that every acquisition rule is compared at equal model capacity.

Table 2: Protocol constants, held fixed across every run.

Parameter	Value
Organism and antibiotic pairs	*E. coli* with ceftriaxone, *K. pneumoniae* with ceftriaxone, *S. aureus* with oxacillin
Features	6 000 binned spectral intensities
Label convention	intermediate and resistant both treated as resistant
Training years / test year	2015 and 2016 (80 % train, 20 % validation) / 2018
Acquisition pool	2017 isolates, divided into $R = 4$ arrival rounds
Classifier	gradient boosted trees (XGBoost), tuned once per pair and then frozen
Target sensitivity τ	0.90, used to set the decision threshold
Conformal level α	0.10, per class (Mondrian) split conformal
Embedding for diversity	50 principal components fitted on the training years
Pseudo label cutoffs (E2)	$p \leq 0.05$ susceptible, $p \geq 0.95$ resistant, weight 0.5
Recalibration share (E5)	25 % of acquired labels withheld from training
Seeds	10 (5 for transfer and for the development search)
Statistical test	paired Wilcoxon signed rank over seeds, Holm correction within each pair

4.4 Evaluation Metrics and Statistical Analysis

Model quality is reported as AUROC on the following year's isolates, with the area under the precision and recall curve and balanced accuracy as secondary measures. Safety is reported in the operational view of Section 3.2, as counts of both error types among the untested isolates. Calibration is reported as realised sensitivity against the 0.90 target, and as per class conformal coverage against the 0.90 guarantee. The primary endpoint, which was specified before any run was executed, was the number of labels required to recover 95 % of the AUROC gap between the two bounds, estimated by bootstrap over seeds. Strategies are compared against random acquisition with a paired Wilcoxon signed rank test over matched seeds, and the Holm procedure is applied within each organism and antibiotic pair in order to control the family wise error rate at 0.05.

5 Results

5.1 Premise: Temporal Decay and the Value of Retraining

A model that is frozen after 2016 loses ground on the 2018 isolates, and full retraining on the 2017 labels recovers it: AUROC rises by 0.044, 0.046 and 0.051 for the three pairs (Fig. 3). The gap is modest, but it is entirely consistent across seeds, and it is this gap that every acquisition strategy is competing to close cheaply. All of the results that follow are expressed relative to these two bounds.

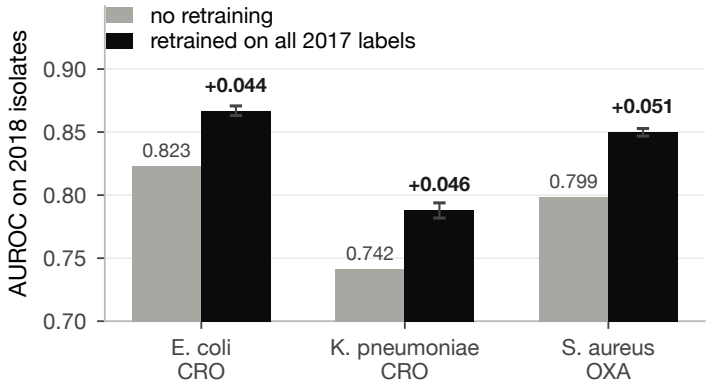


Fig. 3: Retraining recovers accuracy lost to drift. Bars show mean AUROC on 2018 isolates over ten seeds; whiskers are 95 % confidence intervals. Labels above the right hand bars give the gain over not retraining at all.

5.2 Primary Outcome: No Strategy Outperformed Random

Fig. 4 shows the label efficiency curves and Fig. 5 the paired differences against random acquisition. Across the 45 preplanned comparisons, that is three pairs by five budgets by three strategies, **no strategy was significantly better than random once the Holm correction had been applied**. The only significant results run in the opposite direction: uncertainty sampling was worse than random for *K. pneumoniae* at 800 labels (-0.029, Holm $p = 0.029$) and for *S. aureus* at the same budget (-0.013, raw $p = 0.006$, Holm $p = 0.088$). The primary endpoint we had specified in advance was never reached. No strategy recovered 95 % of the retraining gap within 800 labels, and random acquisition itself recovered only 66 to 77 % at that budget. Table 3 gives the values at the final budget.

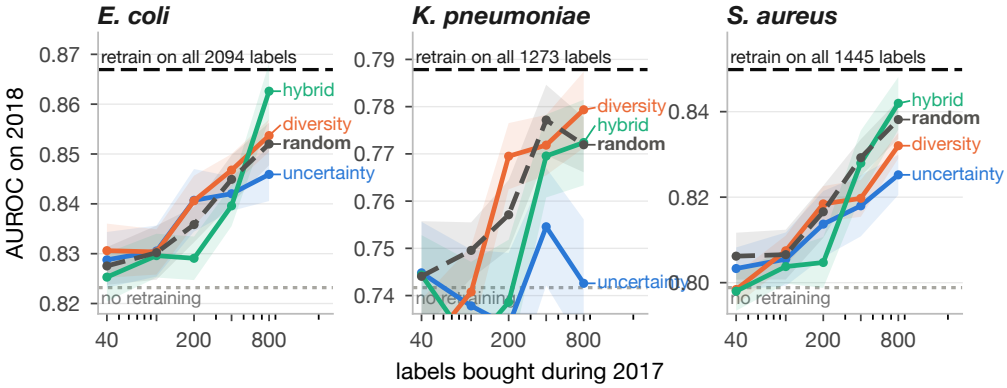


Fig. 4: Label efficiency curves. Mean AUROC on 2018 isolates over ten seeds, with shaded 95 % confidence intervals, against a logarithmic budget axis. The dashed upper line is full retraining, which consumes 1 273 to 2 094 labels, and the dotted lower line is no retraining. All four acquisition rules lie within one another's confidence bands at every budget.

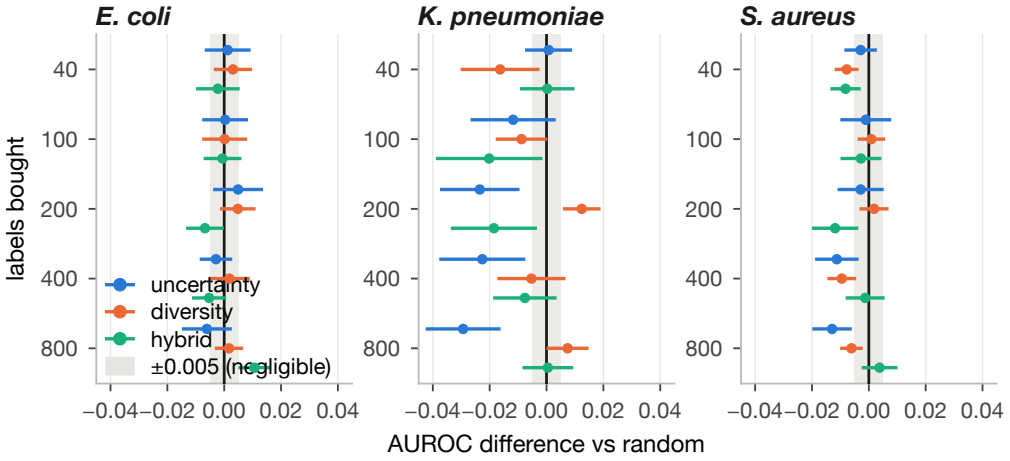


Fig. 5: Paired differences against random acquisition, by pair and budget. Points are mean AUROC differences over ten matched seeds with 95 % confidence intervals; the shaded band marks the difference of ±0.005 prespecified as negligible. No interval excludes zero in a direction favouring the strategy once Holm correction is applied.

Table 3: AUROC on 2018 isolates at the largest budget of 800 labels, mean over ten seeds, with the two bounds. Confidence intervals at 95 % are at most 0.014 throughout and are drawn in Figs. 4 and 5. The best active learning entry per row is set in bold; no difference from random is statistically significant.

Pair	No retraining	Random	Uncertainty	Diversity	Hybrid	Full retraining
E. coli / CRO	0.823	0.852	0.846	0.854	**0.863**	0.867
K. pneumoniae / CRO	0.742	0.772	0.743	**0.779**	0.772	0.788
S. aureus / OXA	0.799	0.838	0.825	0.832	**0.842**	0.850

Because the design is paired over seeds it is reasonably sensitive: with ten seeds it detects an AUROC difference of about 0.008 at 80 % power for the variances we observed. Large gains from these strategies can therefore be excluded in this setting, even though gains of any size cannot. In practical terms, a laboratory that acquires isolates at random gives up nothing measurable in model quality.

5.3 Acquisition Determines Which Patients Are Exposed

The picture changes once the measurement moves from next year's model to this year's patients. Uncertainty sampling spends the budget exactly where the model's call is least trustworthy, so the isolates it sends to the laboratory are disproportionately the ones the model would have got wrong, and those isolates then receive a true phenotype instead of a predicted one. Summed over the year at 800 labels, uncertainty acquisition reduced the resistant isolates that were wrongly reported susceptible by 33, 73 and 51 % relative to random (Fig. 6; all $p < 0.01$, paired Wilcoxon over seeds).

This benefit has to be reported together with its cost on the opposite error, and here the species behave differently (Table 4). Major errors fell by 25 % for *E. coli*, were unchanged for *S. aureus*, and rose by 54 % for *K. pneumoniae*, which is also the pair for which uncertainty acquisition produced the weakest model. What we claim is therefore a reliable reduction in the dangerous error, together with an effect on the other error that depends on the organism, and not a uniform improvement in safety.

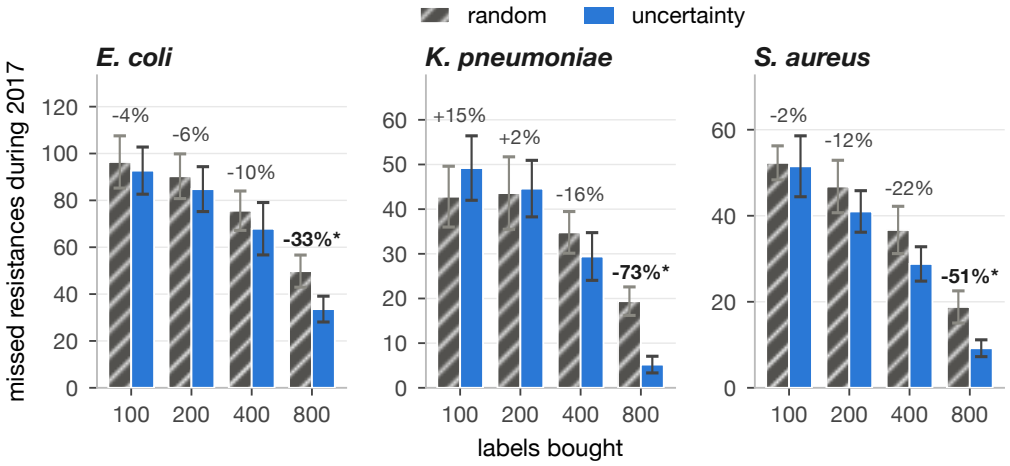


Fig. 6: Resistant isolates wrongly reported susceptible during the year, counted among isolates that were not sent to the laboratory. Bars are means over ten seeds with 95 % confidence intervals; hatched bars are random acquisition and solid bars uncertainty acquisition. Labels give the relative change, with an asterisk marking $p < 0.01$. At small budgets the strategies are indistinguishable; the gap opens as the budget grows.

Table 4: Both error types among untested isolates at a budget of 800 labels, summed over the year, mean over ten seeds. The relative change is uncertainty acquisition against random; p-values are paired Wilcoxon, uncorrected. VME denotes resistant reported susceptible and ME susceptible reported resistant.

Pair	VME random	VME unc.	change	p	ME random	ME unc.	change	p
E. coli	49.8	33.6	−33	0.008	347.8	259.8	−25	0.002
K. pneumoniae	19.4	5.2	−73	0.002	123.7	190.1	+54	0.002
S. aureus	18.8	9.2	−51	0.008	134.6	140.0	+4	0.432

5.4 Both Calibrated Components Had Expired

Both of the calibrated components failed, and neither failure shows up in the discrimination metric that would normally be monitored. The decision threshold had been chosen on the 2015 and 2016 validation isolates so as to catch 90 % of resistant isolates; on the 2018 isolates the same threshold caught 80, 73 and 79 % (Fig. 7a). The conformal rule promises that at most 10 % of resistant isolates will receive a confident susceptible prediction set, and measured over the 2017 isolates it delivered 15.5, 18.9 and 14.3 % (Fig. 7b).

This also explains a result in E3 that is otherwise puzzling. Conformal routing matched random acquisition on AUROC while using fewer labels, between 428 and 650 rather than 800, and yet it left *more* missed resistances than random at the same budget: 78 against 50, 33 against 19 and 39 against 19, with Holm $p < 0.01$ in every case. A routing rule can only be as good as the uncertainty estimate it routes on, and that estimate had quietly expired.

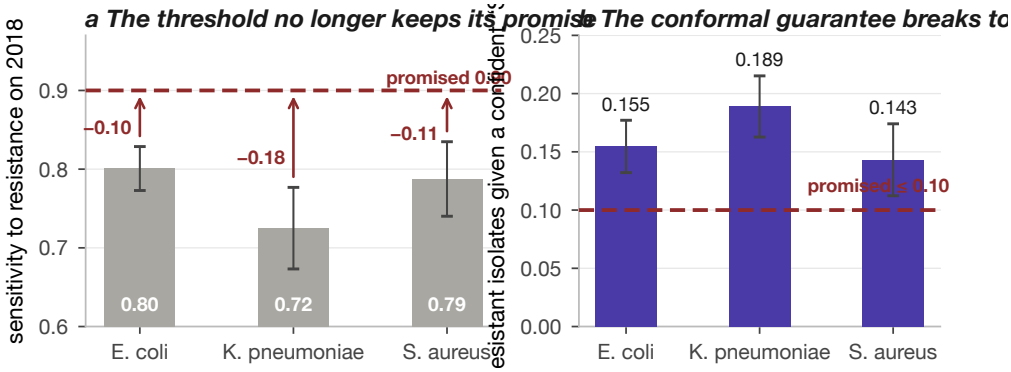


Fig. 7: Two broken promises. (a) The decision threshold was set for 90 % sensitivity to resistance on historical validation data; on 2018 isolates it delivers between 0.10 and 0.18 less. (b) The per class conformal rule promises that at most 10 % of resistant isolates receive a confident susceptible set, and delivers 14 to 19 % over the 2017 isolates. Whiskers are 95 % confidence intervals over seeds.

5.5 Recalibration Restores the Target and Exposes Its Cost

Experiment E5 withholds a quarter of the acquired labels from training and uses them, in each round, to reset both the decision threshold and the conformal bounds. Sensitivity on the 2018 isolates moves back towards the target, from 0.82 to 0.88, from 0.64 to 0.77 and from 0.72 to 0.79 when averaged over strategies and budgets. The major error rate rises correspondingly, from 0.37 to 0.52, from 0.27 to 0.49 and from 0.26 to 0.39, and balanced accuracy falls (Fig. 8). AUROC is essentially unchanged, and marginally lower, because a quarter of the acquired labels no longer contributes to training.

The right reading of this is not that recalibration does harm. It is that the drifted threshold had silently moved to a different operating point, one that looked better on major errors only because it had stopped keeping its sensitivity promise. Restoring the promise shows what 90 % sensitivity really costs on a model whose AUROC lies between 0.75 and 0.85. A laboratory that finds both of those numbers unacceptable needs a better model, not a better threshold.

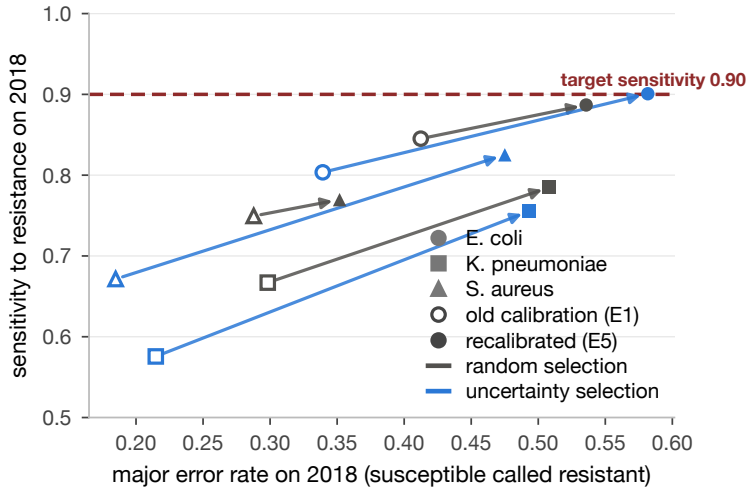


Fig. 8: Recalibration moves every configuration up and to the right: closer to the sensitivity target, at the cost of more susceptible isolates reported resistant. Arrows run from the stale calibration of E1, drawn hollow, to recalibration on freshly acquired labels in E5, drawn filled. Marker shape denotes the species and colour the acquisition rule; each point averages ten seeds and three budgets.

5.6 Adoption at a Different Hospital

The largest effects in the whole study come from transfer, and they have nothing to do with acquisition. A model trained at hospital A and applied unchanged elsewhere ranged from unaffected, for *E. coli* at hospital B with an AUROC of 0.835, to worse than chance, for *K. pneumoniae* at hospital B with an AUROC of 0.477. Local labels repair most of the damage, and they do so quickly: one hundred locally acquired labels moved that worst case from 0.477 to 0.614, which is the full retraining value at that site (Fig. 9). Which isolates were acquired again made no significant difference in any of the 120 comparisons, although the target sites are small and this test has correspondingly little power.

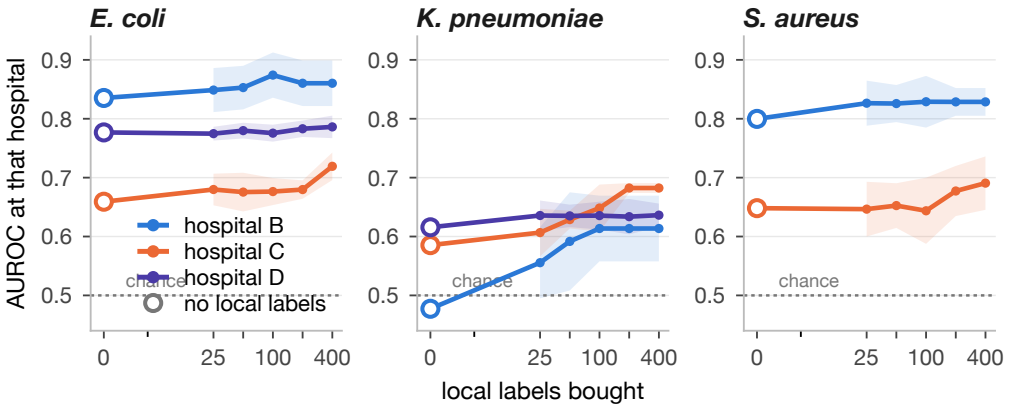


Fig. 9: Adoption of the model at a new hospital. Hollow markers at zero show the transferred model with no local labels; lines show random local acquisition. The collapse for *K. pneumoniae* at hospital B, to below chance, and its repair by 25 to 100 local labels, is the strongest effect observed in the study.

5.7 A Preregistered Search over Method Combinations

A single negative result invites the objection that some better combination would have succeeded. We therefore decomposed the method into five slots, namely the acquisition rule, the split of the budget between targeted and random purchase, the retraining weights, the calibration source, and the use of unlabelled spectra. Enumerating these gave 2 430 combinations. Removing the ones that are incoherent or dominated left 1 424 valid combinations, of which 559 were both robust and affordable, and from those we selected sixteen to execute. One family among them was the family we had expected to win: spend half of the budget on targeted isolates and half at random, then use the random half, which is an unbiased sample of the drifted distribution and is scored before it enters training, to reset the threshold and the conformal bounds.

To keep the search honest, all 510 runs were carried out on a **development split that never touches 2018**. Training used 2015 and 2016, labels were acquired from the first half of the 2017 arrivals, and testing was done on the second half. The rule for selecting winners was written into the version controlled log before any run was executed. As a reproducibility check, 110 of the runs were executed independently on two machines and returned identical metrics.

The preregistered rule selected a combination that reduced missed resistances by 63 % and brought sensitivity up to 0.92. It achieved this by reporting roughly two thirds of susceptible isolates as resistant. Fig. 10 shows the reason: every combination that recalibrates slides along the same error tradeoff rather than improving on it, and our rule had scored only one end of that tradeoff. We report this as a failure of our own decision rule and not as a method result. It is the clearest illustration in the study of how badly a safety metric can mislead when it is read in isolation.

Judged on both error types at once, only one family lies strictly inside the baseline on both axes, and that family is uncertainty acquisition. On its own it gives a ratio of 0.87 on missed resistances ($p = 0.017$) and 0.84 on major errors ($p = 0.002$), at an AUROC difference of $+0.002$. Mixed half and half with random it gives ratios of 0.95 ($p = 0.16$) and 0.89 ($p = 0.006$) at -0.001 AUROC. No combination of recalibration, recency weighting, importance weighting, drift novelty acquisition or conformal pseudo labelling improved on it. Mean AUROC across all sixteen combinations spanned 0.815 to 0.839, so model quality was flat whatever the configuration.

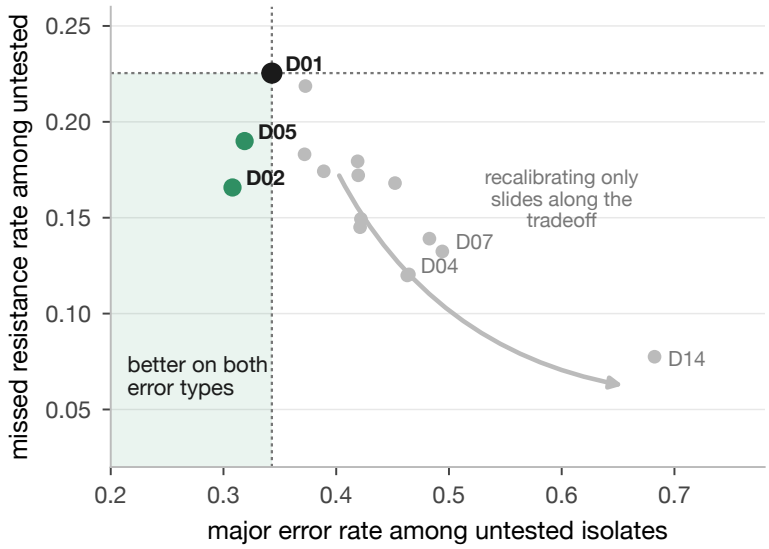


Fig. 10: The error tradeoff on the development split. Each point is one of sixteen method combinations, averaged over three pairs, two budgets and five seeds. Only points in the shaded quadrant improve on the random baseline D01 on both error types. The combination selected by our preregistered rule, D14, lies far to the right: it purchases a low missed resistance rate by reporting most susceptible isolates as resistant.

5.8 An Invalid Experiment: Pseudo labelling

Experiment E2 added self training. Isolates that the model was confident about received pseudo labels, with $p \leq 0.05$ treated as susceptible and $p \geq 0.95$ as resistant, and were then added to the training set at a reduced weight. The results were poor. Pseudo labels on their own fell below no retraining at all, and major errors rose sharply at the smaller budgets.

They are also uninterpretable, and the reason is an error in our own design. The decision threshold for these models lies between 0.010 and 0.030, that is, **below the pseudo susceptible cutoff of 0.05**. Isolates falling in that band were labelled susceptible for training at the same time as the deployed model was reporting them as resistant (Fig. 11). The procedure was therefore injecting systematically contradictory labels: between 62 and 66 % of each round's arrivals were pseudo labelled, at an accuracy of 89 to 92 %, which works out at 60 to 240 incorrect labels per run. We report the experiment as invalid and leave the semisupervised question open, rather than publish a negative result that was caused by our own implementation. A corrected design would tie the cutoffs to the operating threshold, or better still derive them from recalibrated conformal sets.

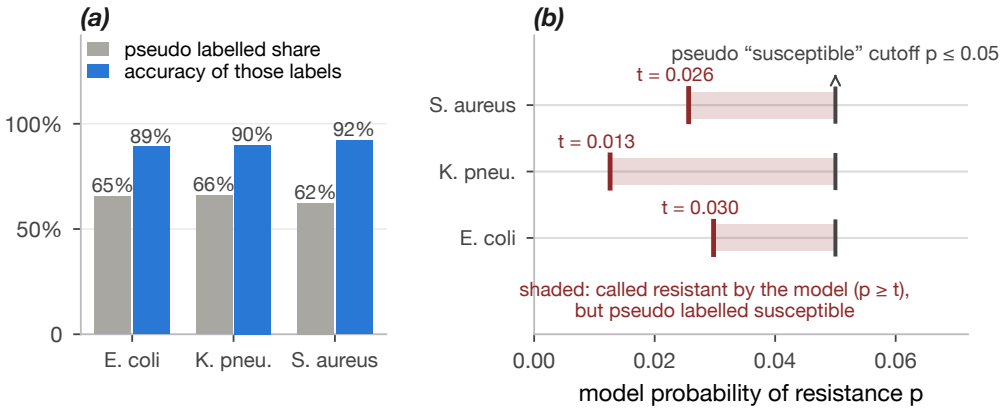


Fig. 11: Why the pseudo labelling experiment is not a fair test. (a) Most arrivals received a pseudo label, roughly one in ten of which was wrong. (b) The fixed cutoff $p \leq 0.05$ for "susceptible" lies above the deployed decision threshold t for all three pairs, so every isolate in the shaded band was trained on as susceptible while being reported as resistant.

6 Discussion

6.1 Label Efficiency Is the Wrong Objective Here

Active learning is conventionally evaluated by asking how few labels are needed to reach a target accuracy. On this task that number turns out to be dominated by factors the acquisition rule has no control over: the signal that is available in a 6 000 bin spectrum, the prevalence of the resistant class, and the size of the drift between years. Our strategies moved it by less than the variation between seeds. The very same choice, however, moved the number of patients who received a wrong susceptible report by as much as a factor of four. For a clinical deployment it is the second quantity that is worth optimising, and it is invisible to the standard evaluation.

6.2 Calibration Is a Maintained Component, Not a Training Artefact

Both of the calibrated components in our system were fitted once and then trusted for two years, which is exactly what a deployed system does by default. Both failed quietly. The threshold lost between 10 and 18 points of sensitivity, and the conformal rule exceeded its own error budget by a factor of 1.4 to 1.9, while AUROC, which is the number that would ordinarily be monitored, barely moved at all. Any deployment of this kind of model needs a stream of freshly phenotyped isolates set aside for calibration, and our results point to an inexpensive way of obtaining one: a random share of the testing budget is an unbiased sample of the current distribution and can be scored before it enters training. In our experiments this did not improve the error tradeoff. It remains worth doing all the same, because it is what keeps a stated guarantee true.

6.3 Practical Recommendations

The findings translate into four concrete recommendations for a laboratory that maintains such a model.

1. Acquire isolates for retraining by whatever means is operationally simplest. Random acquisition is not worse in model quality.
2. Where the budget allows, bias acquisition towards the isolates the model is uncertain about, not for the model's benefit, but because those are the isolates whose incorrect calls would otherwise reach patients. The effect on the number of susceptible isolates reported as resistant should be expected to vary by organism, and it should be measured rather than assumed.
3. Recalibrate the decision threshold and any uncertainty sets on recent labels, and treat a drifting realised sensitivity as an alarm in its own right.
4. When adopting a model that was developed at another hospital, acquire 25 to 100 local labels before relying on it. This was the single most valuable intervention we measured.

7 Limitations

There are several limitations which qualify these conclusions.

- **No acquisition dates.** DRIAMS does not provide them, so the arrival order within the year has to be simulated by a hash based partition. Seasonal structure and outbreak dynamics are therefore absent, and the drift within the year that would most favour adaptive acquisition cannot be tested here. This is the limitation we would most like to remove, and removing it requires a dataset with timestamps.
- **Three pairs, four hospitals, one source.** The organisms with enough labelled isolates for this design are few, and all of the data originate from one country and one collection effort.
- **Simulation rather than deployment.** Labels are revealed from an existing database instead of being produced by a laboratory, so turnaround time, the batch effects of repeating a test in practice, and clinician response are all out of scope.
- **Statistical power.** Ten seeds support the detection of AUROC differences of about 0.008; the transfer tests, with five seeds and small target sites, are weaker. Absence of evidence here is not strong evidence of absence for small effects.
- **One classifier family.** All of the results use tuned gradient boosted trees. A representation with better behaved predictive uncertainty could change the ranking of the acquisition rules, and uncertainty sampling in particular.
- **The semisupervised question remains open** by our own admission, for the reason given in Section 5.8.

8 Conclusion

Across 2 859 retraining simulations on DRIAMS, no active learning strategy produced a better resistance model than simply acquiring labels at random, and the standard label efficiency endpoint was never reached. The same experiments nevertheless show that the choice is consequential. Acquiring the uncertain isolates reduced the resistant isolates that were wrongly reported susceptible by between a third and three quarters, while the decision threshold and the conformal guarantees that were supposed to bound such errors had both quietly expired after two years of drift. A preregistered search over sixteen method combinations, executed on a development split that never touched the test year, reproduced these conclusions, and by selecting a combination that merely exchanged one error for the other it illustrated how readily a safety metric misleads when it is read in isolation. We recommend that acquisition policies for this task be reported with counts of both error types at deployment, and that calibration be treated as a maintained component rather than a property of the training run.

Reproducibility. Every experiment is driven by a single configuration file, and each run writes a self describing record containing its seed, a fingerprint of the data split, library versions and per round metrics. 110 runs were executed independently on two machines and returned identical metrics. Code and result records will be released on publication.

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